

PYS-01 · BIOMEDICAL

Magnetic Capsule Maze Cup

Steer a magnetic pill-camera through a body maze without touching it.

PREDICT > BUILD or SOLVE > TEST > SCORE > REDESIGN

WIN: Fastest clean run with low penalties and a clear magnetic-control explanation.

PYS-02 · MATERIALS

Oobleck Armor Arena

Design armor with a material that acts like a liquid until impact.

PREDICT > BUILD or SOLVE > TEST > SCORE > REDESIGN

WIN: Best target protection with least material and cleanest reset.

PYS-03 · MECHANICS

Cardboard Automata Arcade

Build a one-crank machine that performs a mini task.

PREDICT > **BUILD or SOLVE** > **TEST** > **SCORE** > **REDESIGN**

WIN: Most reliable motion with useful mechanism and clear explanation.

PYS-04 · BIOMEDICAL

Stethoscope Sprint and Recovery

Build a medical listening tool and use it like a sports scientist.

PREDICT > BUILD or SOLVE > TEST > SCORE > REDESIGN

WIN: Best sound quality plus accurate, respectful data collection.

PYS-05 · BIOMEDICAL

Reaction Time Combine

Train like a sports scientist and prove which strategy cuts delay.

PREDICT > BUILD or SOLVE > TEST > SCORE > REDESIGN

WIN: Best median plus most improvement using clean data.

PYS-06 · WAVES + SOUND

SONAR Slinky Showdown

Make invisible distance sensing visible with a slinky pulse.

PREDICT > BUILD or SOLVE > TEST > SCORE > REDESIGN

WIN: Most correct wave predictions and station solutions.

PYS-07 · OPTICS + LIGHT

Pinhole Precision Challenge

Build the sharpest camera image with no lens.

PREDICT > BUILD or SOLVE > TEST > SCORE > REDESIGN

WIN: Best image clarity and concept explanation after one aperture redesign.

PYS-08 · MECHANICS

Low-Ropes Force Map Relay

Predict balance and stability before the ropes element proves you right.

PREDICT > **BUILD or SOLVE** > **TEST** > **SCORE** > **REDESIGN**

WIN: Most accurate predictions and clear debrief connections.

PYS-09 · MECHANICS

Hovercraft Hockey Hackathon

Build a puck that moves by barely touching the floor.

PREDICT > BUILD or SOLVE > TEST > SCORE > REDESIGN

WIN: Best glide plus control in a gym-safe tournament.

PYS-10 · OPTICS + LIGHT

Spectra Sleuth Showdown

Can two white lights hide different color fingerprints?

PREDICT > BUILD or SOLVE > TEST > SCORE > REDESIGN

WIN: Most correct spectrum matches and strongest safe fireworks-science explanation.

PYS-11 · COMPUTING

BookBot Bin Logic Challenge

Retrieve the right book by using bin addresses, not a conveyor race.

PREDICT > BUILD or SOLVE > TEST > SCORE > REDESIGN

WIN: Most correct retrievals with few traffic conflicts and clear algorithm defense.

PYS-12 · DESIGN

Accessibility Ramp Rescue Lab

Design a portable ramp that works for a real user constraint.

PREDICT > BUILD or SOLVE > TEST > SCORE > REDESIGN

WIN: Best safe ramp meeting slope, load, portability, and client criteria.

PYB-01 · MECHANICS

Pencil Pressure Safety Lab

Spread a force before it dents the target.

PREDICT > BUILD or SOLVE > TEST > SCORE > REDESIGN

WIN: Largest area protected with clear data.

PYB-02 · BIOMEDICAL

Domino Neuron Relay

Make a nerve signal survive a gap.

PREDICT > BUILD or SOLVE > TEST > SCORE > REDESIGN

WIN: Fastest reliable signal with best explanation.

PYB-03 · MECHANICS

Pulley Rescue Relay

Lift the load with smarter force, not stronger arms.

PREDICT > BUILD or SOLVE > TEST > SCORE > REDESIGN

WIN: Best lift with correct setup and explanation.

PYB-04 · COMPUTING

Barcode Checksum Rescue

Catch the fake barcode before the warehouse ships the wrong part.

PREDICT > BUILD or SOLVE > TEST > SCORE > REDESIGN

WIN: Most bad codes detected with few false alarms.